Array Design: Full Candidate Report

All 274 (Aperture, Pitch, Packing) Designs – Pass/Fail, Metrics, and Reason

Source: `Array_Design/array_optimization/stage3_decision/array_candidates.csv`

Methodology: how PASS/FAIL is determined

Every row is one (hole diameter N, pitch s, packing) design, evaluated across every tested myopia level M and reported at its **worst case across** M. Two independent conditions must both hold for a design to be `Safe`:

1. **Ghosting risk** < 1.0 – the continuous risk model ($P(N_{\text{eff}} \geq 2)\times$worst-harmonic perceptual margin, Barten-1999-CSF-based), not the deprecated binary Regime A/B/C gate. Risk $= 1.0$ is the point where the probability-weighted ghost comb crosses the eye's own contrast-detection threshold.
2. **Pitch margin** ≥ 1.0 – pitch is at or beyond its own no-ghost geometric ceiling (s_{solo}, the pitch above which a second hole can *never* be simultaneously active, for that packing). This is a structural (geometry-only) guarantee, not a statistical one.

A design that is `Safe` is then either **PASS – Pareto-optimal** (`recommended_for_prototyping=True`: on the Pareto front of blur vs. whole-lens transmission, among safe designs) or **PASS – dominated** (safe, but another safe design is at least as good on both blur and transmission, and strictly better on one – still physically safe, just not on the recommended frontier). A design that fails either condition is `FAIL`, broken down by which condition(s) failed. **No numeric transmission target (OD/%T) is enforced anywhere** – T is reported for every row but is not itself a pass/fail criterion (see this project's own `GAPS_CLOSED.md` Gap 4).

Column key: N = hole diameter (μm); s = pitch (μm); T = whole-lens transmission (%); blur = $r_{EE}(50\%)$ (μm, worst-case across M; historically named `r_EE_80_um` in the source CSV for schema reasons, see `decide.py`'s own docstring); risk = worst-case-across-M continuous ghosting risk; margin = pitch as a fraction of its own no-ghost ceiling (≥ 1.0 required to pass).

Summary

Category	Count
Total candidate designs	274
Safe (risk < 1.0 **and** margin ≥ 1.0)	90
PASS – Pareto-optimal (`recommended_for_prototyping`)	**20**
PASS – safe but dominated (not on the Pareto front)	70
Fail	184
FAIL – ghosting risk only	0
FAIL – pitch margin only	2
FAIL – both risk and margin	182

Most failing designs fail *both* conditions simultaneously, not one or the other: a pitch too dense to clear its own no-ghost ceiling is, for the same geometric reason, also the pitch regime where the continuous risk model reports real ghosting risk. The table below is sorted PASS–Pareto first, then PASS–dominated, then FAIL, and by blur ascending within each group.

Full candidate table

Table 1: All 274 array candidates, pass/fail, metrics, and reason.

N	s	pattern	$T(\%)$	blur(μm)	risk	margin	Safe	Rec	Reason
400	4700	square	0.57	17.3	0.0	1.19	Yes	Yes	PASS – Pareto-optimal
400	4200	hex	0.82	17.4	0.0	1.05	Yes	Yes	PASS – Pareto-optimal
500	4700	square	0.89	23.5	0.0	1.19	Yes	Yes	PASS – Pareto-optimal
500	4200	square	1.11	23.5	0.0	1.07	Yes	Yes	PASS – Pareto-optimal
500	4200	hex	1.29	23.5	0.0	1.05	Yes	Yes	PASS – Pareto-optimal
600	4200	hex	1.85	26.7	0.0	1.05	Yes	Yes	PASS – Pareto-optimal
700	4700	hex	2.01	32.6	0.0	1.17	Yes	Yes	PASS – Pareto-optimal
700	4200	hex	2.52	32.6	0.0	1.05	Yes	Yes	PASS – Pareto-optimal
750	4200	hex	2.89	34.9	0.0	1.05	Yes	Yes	PASS – Pareto-optimal
1000	5200	square	2.90	49.7	0.0	1.32	Yes	Yes	PASS – Pareto-optimal
1000	5200	hex	3.35	49.7	0.0	1.30	Yes	Yes	PASS – Pareto-optimal
1000	4200	square	4.45	49.7	0.0	1.07	Yes	Yes	PASS – Pareto-optimal
1000	4200	hex	5.14	49.8	0.0	1.05	Yes	Yes	PASS – Pareto-optimal
1250	5200	hex	5.24	64.8	0.0	1.30	Yes	Yes	PASS – Pareto-optimal
1250	4200	hex	8.03	64.8	0.0	1.05	Yes	Yes	PASS – Pareto-optimal
1500	4700	hex	9.24	80.1	0.0	1.17	Yes	Yes	PASS – Pareto-optimal
1500	4200	hex	11.57	80.2	0.0	1.05	Yes	Yes	PASS – Pareto-optimal
1750	4200	hex	15.74	95.3	0.0	1.05	Yes	Yes	PASS – Pareto-optimal
2000	4700	hex	16.42	109.5	0.0	1.17	Yes	Yes	PASS – Pareto-optimal
2000	4200	hex	20.56	109.5	0.0	1.05	Yes	Yes	PASS – Pareto-optimal
400	4700	hex	0.66	17.4	0.0	1.17	Yes	No	PASS – dominated
400	5200	hex	0.54	17.4	0.0	1.30	Yes	No	PASS – dominated
400	4200	square	0.71	17.4	0.0	1.07	Yes	No	PASS – dominated
400	5200	square	0.46	17.4	0.0	1.32	Yes	No	PASS – dominated
300	5200	hex	0.30	18.5	0.0	1.30	Yes	No	PASS – dominated
300	5200	square	0.26	18.5	0.0	1.32	Yes	No	PASS – dominated
300	4200	square	0.40	18.5	0.0	1.07	Yes	No	PASS – dominated
300	4700	hex	0.37	18.5	0.0	1.17	Yes	No	PASS – dominated
300	4200	hex	0.46	18.5	0.0	1.05	Yes	No	PASS – dominated
300	4700	square	0.32	18.5	0.0	1.19	Yes	No	PASS – dominated

(continued on next page)

(continued from previous page)

N	s	pattern	T(%)	blur(μm)	risk	margin	Safe	Rec	Reason
500	5200	square	0.73	23.5	0.0	1.32	Yes	No	PASS – dominated
500	5200	hex	0.84	23.5	0.0	1.30	Yes	No	PASS – dominated
500	4700	hex	1.03	23.5	0.0	1.17	Yes	No	PASS – dominated
200	4700	square	0.14	25.7	0.0	1.19	Yes	No	PASS – dominated
200	4200	hex	0.21	25.7	0.0	1.05	Yes	No	PASS – dominated
200	4700	hex	0.16	25.7	0.0	1.17	Yes	No	PASS – dominated
200	4200	square	0.18	25.7	0.0	1.07	Yes	No	PASS – dominated
200	5200	hex	0.13	25.7	0.0	1.30	Yes	No	PASS – dominated
200	5200	square	0.12	25.7	0.0	1.32	Yes	No	PASS – dominated
600	5200	square	1.05	26.6	0.0	1.32	Yes	No	PASS – dominated
600	5200	hex	1.21	26.7	0.0	1.30	Yes	No	PASS – dominated
600	4700	square	1.28	26.7	0.0	1.19	Yes	No	PASS – dominated
600	4700	hex	1.48	26.7	0.0	1.17	Yes	No	PASS – dominated
600	4200	square	1.60	26.7	0.0	1.07	Yes	No	PASS – dominated
700	4700	square	1.74	32.6	0.0	1.19	Yes	No	PASS – dominated
700	4200	square	2.18	32.6	0.0	1.07	Yes	No	PASS – dominated
700	5200	hex	1.64	32.6	0.0	1.30	Yes	No	PASS – dominated
700	5200	square	1.42	32.6	0.0	1.32	Yes	No	PASS – dominated
150	5200	hex	0.08	33.9	0.0	1.30	Yes	No	PASS – dominated
150	4200	hex	0.12	33.9	0.0	1.05	Yes	No	PASS – dominated
150	5200	square	0.07	33.9	0.0	1.32	Yes	No	PASS – dominated
150	4700	hex	0.09	33.9	0.0	1.17	Yes	No	PASS – dominated
150	4700	square	0.08	33.9	0.0	1.19	Yes	No	PASS – dominated
150	4200	square	0.10	33.9	0.0	1.07	Yes	No	PASS – dominated
750	4700	square	2.00	34.9	0.0	1.19	Yes	No	PASS – dominated
750	5200	square	1.63	34.9	0.0	1.32	Yes	No	PASS – dominated
750	5200	hex	1.89	34.9	0.0	1.30	Yes	No	PASS – dominated
750	4700	hex	2.31	34.9	0.0	1.17	Yes	No	PASS – dominated
750	4200	square	2.50	34.9	0.0	1.07	Yes	No	PASS – dominated
1000	4700	hex	4.11	49.8	0.0	1.17	Yes	No	PASS – dominated
1000	4700	square	3.56	49.8	0.0	1.19	Yes	No	PASS – dominated
100	5200	hex	0.03	50.5	0.0	1.30	Yes	No	PASS – dominated
100	4200	hex	0.05	50.5	0.0	1.05	Yes	No	PASS – dominated
100	5200	square	0.03	50.5	0.0	1.32	Yes	No	PASS – dominated
100	4700	hex	0.04	50.5	0.0	1.17	Yes	No	PASS – dominated
100	4200	square	0.04	50.5	0.0	1.07	Yes	No	PASS – dominated
100	4700	square	0.04	50.5	0.0	1.19	Yes	No	PASS – dominated
1250	5200	square	4.54	64.8	0.0	1.32	Yes	No	PASS – dominated
1250	4700	hex	6.41	64.8	0.0	1.17	Yes	No	PASS – dominated

(continued on next page)

(continued from previous page)

N	s	pattern	T(%)	blur(μm)	risk	margin	Safe	Rec	Reason
1250	4200	square	6.96	64.8	0.0	1.07	Yes	No	PASS – dominated
1250	4700	square	5.56	64.8	0.0	1.19	Yes	No	PASS – dominated
75	5200	hex	0.02	67.2	0.0	1.30	Yes	No	PASS – dominated
75	4700	hex	0.02	67.2	0.0	1.17	Yes	No	PASS – dominated
75	5200	square	0.02	67.3	0.0	1.32	Yes	No	PASS – dominated
75	4700	square	0.02	67.3	0.0	1.19	Yes	No	PASS – dominated
75	4200	hex	0.03	67.3	0.0	1.05	Yes	No	PASS – dominated
75	4200	square	0.03	67.3	0.0	1.07	Yes	No	PASS – dominated
1500	5200	square	6.54	80.1	0.0	1.32	Yes	No	PASS – dominated
1500	4700	square	8.00	80.1	0.0	1.19	Yes	No	PASS – dominated
1500	5200	hex	7.55	80.1	0.0	1.30	Yes	No	PASS – dominated
1500	4200	square	10.02	80.2	0.0	1.07	Yes	No	PASS – dominated
1750	5200	square	8.90	95.3	0.0	1.32	Yes	No	PASS – dominated
1750	5200	hex	10.27	95.3	0.0	1.30	Yes	No	PASS – dominated
1750	4700	square	10.89	95.3	0.0	1.19	Yes	No	PASS – dominated
1750	4700	hex	12.57	95.3	0.0	1.17	Yes	No	PASS – dominated
1750	4200	square	13.64	95.3	0.0	1.07	Yes	No	PASS – dominated
2000	5200	square	11.62	109.5	0.0	1.32	Yes	No	PASS – dominated
2000	5200	hex	13.42	109.5	0.0	1.30	Yes	No	PASS – dominated
2000	4700	square	14.22	109.5	0.0	1.19	Yes	No	PASS – dominated
2000	4200	square	17.81	109.5	0.0	1.07	Yes	No	PASS – dominated
400	3700	square	0.92	17.4	16.2	0.94	No	No	FAIL – risk & margin
300	3700	square	0.52	18.6	14.9	0.94	No	No	FAIL – risk & margin
500	3700	square	1.43	23.5	17.0	0.94	No	No	FAIL – risk & margin
200	3700	square	0.23	25.7	13.3	0.94	No	No	FAIL – risk & margin
600	3700	square	2.07	26.7	17.5	0.94	No	No	FAIL – risk & margin
700	3700	square	2.81	32.6	17.7	0.94	No	No	FAIL – risk & margin
150	3700	square	0.13	33.9	11.8	0.94	No	No	FAIL – risk & margin
750	3700	square	3.23	34.9	17.8	0.94	No	No	FAIL – risk & margin
400	3700	hex	1.06	39.7	30.2	0.92	No	No	FAIL – risk & margin
300	3700	hex	0.60	40.7	27.9	0.92	No	No	FAIL – risk & margin
500	3700	hex	1.66	45.3	31.8	0.92	No	No	FAIL – risk & margin
200	3700	hex	0.26	47.4	24.9	0.92	No	No	FAIL – risk & margin
600	3700	hex	2.38	48.3	32.7	0.92	No	No	FAIL – risk & margin
1000	3700	square	5.74	49.8	17.8	0.94	No	No	FAIL – risk & margin
100	3700	square	0.06	50.5	9.6	0.94	No	No	FAIL – risk & margin
700	3700	hex	3.25	53.7	33.1	0.92	No	No	FAIL – risk & margin
150	3700	hex	0.15	55.0	22.0	0.92	No	No	FAIL – risk & margin
750	3700	hex	3.73	55.9	33.3	0.92	No	No	FAIL – risk & margin

(continued on next page)

(continued from previous page)

N	s	pattern	T(%)	blur(μm)	risk	margin	Safe	Rec	Reason
1250	3700	square	8.96	64.8	17.2	0.94	No	No	FAIL – risk & margin
75	3700	square	0.03	67.3	7.7	0.94	No	No	FAIL – risk & margin
1000	3700	hex	6.62	69.6	33.2	0.92	No	No	FAIL – risk & margin
100	3700	hex	0.07	70.4	17.9	0.92	No	No	FAIL – risk & margin
400	3200	square	1.23	73.5	99.7	0.81	No	No	FAIL – risk & margin
300	3200	square	0.69	74.4	95.7	0.81	No	No	FAIL – risk & margin
500	3200	square	1.92	78.3	105.4	0.81	No	No	FAIL – risk & margin
200	3200	square	0.31	80.0	84.4	0.81	No	No	FAIL – risk & margin
1500	3700	square	12.91	80.1	16.4	0.94	No	No	FAIL – risk & margin
600	3200	square	2.76	80.7	108.9	0.81	No	No	FAIL – risk & margin
1250	3700	hex	10.35	83.5	32.2	0.92	No	No	FAIL – risk & margin
700	3200	square	3.76	85.4	110.7	0.81	No	No	FAIL – risk & margin
75	3700	hex	0.04	86.1	14.4	0.92	No	No	FAIL – risk & margin
150	3200	square	0.17	86.6	76.0	0.81	No	No	FAIL – risk & margin
750	3200	square	4.31	87.2	111.2	0.81	No	No	FAIL – risk & margin
1750	3700	square	17.57	95.3	15.3	0.94	No	No	FAIL – risk & margin
1500	3700	hex	14.91	97.8	30.6	0.92	No	No	FAIL – risk & margin
1000	3200	square	7.67	98.8	110.2	0.81	No	No	FAIL – risk & margin
100	3200	square	0.08	99.9	59.2	0.81	No	No	FAIL – risk & margin
400	3200	hex	1.42	107.0	142.9	0.80	No	No	FAIL – risk & margin
300	3200	hex	0.80	107.9	137.2	0.80	No	No	FAIL – risk & margin
2000	3700	square	22.95	109.5	14.0	0.94	No	No	FAIL – risk & margin
1250	3200	square	11.98	110.7	105.5	0.81	No	No	FAIL – risk & margin
500	3200	hex	2.21	110.9	151.1	0.80	No	No	FAIL – risk & margin
1750	3700	hex	20.29	111.9	28.6	0.92	No	No	FAIL – risk & margin
200	3200	hex	0.35	112.6	121.0	0.80	No	No	FAIL – risk & margin
600	3200	hex	3.19	113.0	156.1	0.80	No	No	FAIL – risk & margin
75	3200	square	0.04	113.7	49.6	0.81	No	No	FAIL – risk & margin
700	3200	hex	4.34	117.0	158.6	0.80	No	No	FAIL – risk & margin
150	3200	hex	0.20	118.1	108.9	0.80	No	No	FAIL – risk & margin
750	3200	hex	4.98	118.4	159.3	0.80	No	No	FAIL – risk & margin
1500	3200	square	17.26	123.0	98.3	0.81	No	No	FAIL – risk & margin
2000	3700	hex	26.50	125.2	26.1	0.92	No	No	FAIL – risk & margin
1000	3200	hex	8.86	128.2	157.9	0.80	No	No	FAIL – risk & margin
100	3200	hex	0.09	129.5	84.9	0.80	No	No	FAIL – risk & margin
1750	3200	square	23.49	135.2	89.3	0.81	No	No	FAIL – risk & margin
1250	3200	hex	13.84	138.2	151.2	0.80	No	No	FAIL – risk & margin
75	3200	hex	0.05	141.5	71.0	0.80	No	No	FAIL – risk & margin
2000	3200	square	30.68	146.8	78.8	0.81	No	No	FAIL – risk & margin

(continued on next page)

(continued from previous page)

N	s	pattern	T(%)	blur(μm)	risk	margin	Safe	Rec	Reason
1500	3200	hex	19.93	148.7	140.9	0.80	No	No	FAIL – risk & margin
400	2700	hex	1.99	158.7	266.7	0.67	No	No	FAIL – risk & margin
300	2700	hex	1.12	159.2	253.2	0.67	No	No	FAIL – risk & margin
1750	3200	hex	27.12	159.2	127.9	0.80	No	No	FAIL – risk & margin
500	2700	hex	3.11	160.7	272.6	0.67	No	No	FAIL – risk & margin
600	2700	hex	4.48	162.0	274.2	0.67	No	No	FAIL – risk & margin
200	2700	hex	0.50	162.1	225.1	0.67	No	No	FAIL – risk & margin
700	2700	hex	6.10	164.5	277.8	0.67	No	No	FAIL – risk & margin
750	2700	hex	7.00	165.3	278.8	0.67	No	No	FAIL – risk & margin
150	2700	hex	0.28	165.5	197.0	0.67	No	No	FAIL – risk & margin
400	2700	square	1.72	165.6	265.9	0.69	No	No	FAIL – risk & margin
300	2700	square	0.97	166.0	252.5	0.69	No	No	FAIL – risk & margin
500	2700	square	2.69	167.1	271.8	0.69	No	No	FAIL – risk & margin
600	2700	square	3.88	168.0	273.4	0.69	No	No	FAIL – risk & margin
200	2700	square	0.43	168.3	224.4	0.69	No	No	FAIL – risk & margin
2000	3200	hex	35.43	169.3	112.9	0.80	No	No	FAIL – risk & margin
700	2700	square	5.28	170.1	277.0	0.69	No	No	FAIL – risk & margin
750	2700	square	6.06	170.7	278.0	0.69	No	No	FAIL – risk & margin
150	2700	square	0.24	171.2	196.4	0.69	No	No	FAIL – risk & margin
1000	2700	hex	12.44	171.2	273.5	0.67	No	No	FAIL – risk & margin
100	2700	hex	0.12	173.1	158.9	0.67	No	No	FAIL – risk & margin
1000	2700	square	10.77	175.3	272.7	0.69	No	No	FAIL – risk & margin
1250	2700	hex	19.44	177.4	256.2	0.67	No	No	FAIL – risk & margin
100	2700	square	0.11	177.8	158.5	0.69	No	No	FAIL – risk & margin
1250	2700	square	16.83	180.6	255.5	0.69	No	No	FAIL – risk & margin
75	2700	hex	0.07	181.6	132.2	0.67	No	No	FAIL – risk & margin
1500	2700	hex	27.99	184.2	230.9	0.67	No	No	FAIL – risk & margin
75	2700	square	0.06	185.4	131.8	0.69	No	No	FAIL – risk & margin
1500	2700	square	24.24	186.8	230.2	0.69	No	No	FAIL – risk & margin
1750	2700	hex	38.10	191.3	200.0	0.67	No	No	FAIL – risk & margin
1750	2700	square	32.99	193.5	199.4	0.69	No	No	FAIL – risk & margin
500	2200	hex	4.68	193.9	385.4	0.55	No	No	FAIL – risk & margin
400	2200	hex	3.00	194.0	375.9	0.55	No	No	FAIL – risk & margin
600	2200	hex	6.75	194.1	386.9	0.55	No	No	FAIL – risk & margin
300	2200	hex	1.69	194.7	352.3	0.55	No	No	FAIL – risk & margin
700	2200	hex	9.18	194.8	382.8	0.55	No	No	FAIL – risk & margin
750	2200	hex	10.54	195.0	379.2	0.55	No	No	FAIL – risk & margin
1000	2200	hex	18.74	197.0	349.0	0.55	No	No	FAIL – risk & margin
200	2200	hex	0.75	197.1	306.8	0.55	No	No	FAIL – risk & margin

(continued on next page)

(continued from previous page)

N	s	pattern	T(%)	blur(μm)	risk	margin	Safe	Rec	Reason
2000	2700	hex	49.76	198.5	165.5	0.67	No	No	FAIL – risk & margin
150	2200	hex	0.42	198.6	275.4	0.55	No	No	FAIL – risk & margin
1250	2200	hex	29.28	199.9	304.7	0.55	No	No	FAIL – risk & margin
2000	2700	square	43.09	200.3	165.1	0.69	No	No	FAIL – risk & margin
100	2200	hex	0.19	201.3	222.9	0.55	No	No	FAIL – risk & margin
1500	2200	hex	42.16	204.1	254.3	0.55	No	No	FAIL – risk & margin
1000	2200	square	16.23	204.3	354.3	0.56	No	No	FAIL – risk & margin
750	2200	square	9.13	205.4	384.9	0.56	No	No	FAIL – risk & margin
1250	2200	square	25.36	205.7	309.3	0.56	No	No	FAIL – risk & margin
700	2200	square	7.95	206.0	388.6	0.56	No	No	FAIL – risk & margin
75	2200	hex	0.11	206.6	183.4	0.55	No	No	FAIL – risk & margin
600	2200	square	5.84	206.7	392.7	0.56	No	No	FAIL – risk & margin
500	2200	square	4.06	208.1	391.3	0.56	No	No	FAIL – risk & margin
200	2200	square	0.65	208.2	311.5	0.56	No	No	FAIL – risk & margin
150	2200	square	0.37	208.3	279.6	0.56	No	No	FAIL – risk & margin
1750	2200	hex	57.38	208.6	200.5	0.55	No	No	FAIL – risk & margin
300	2200	square	1.46	208.7	357.6	0.56	No	No	FAIL – risk & margin
400	2200	square	2.60	208.8	381.6	0.56	No	No	FAIL – risk & margin
1500	2200	square	36.51	209.2	258.2	0.56	No	No	FAIL – risk & margin
100	2200	square	0.16	209.6	226.3	0.56	No	No	FAIL – risk & margin
750	1700	square	15.29	210.8	373.5	0.43	No	No	FAIL – risk & margin
600	1700	square	9.78	210.8	389.6	0.43	No	No	FAIL – risk & margin
700	1700	square	13.32	210.8	380.4	0.43	No	No	FAIL – risk & margin
500	1700	square	6.79	211.0	390.3	0.43	No	No	FAIL – risk & margin
1000	1700	square	27.18	211.1	320.9	0.43	No	No	FAIL – risk & margin
400	1700	square	4.35	211.3	382.2	0.43	No	No	FAIL – risk & margin
300	1700	square	2.45	211.3	367.7	0.43	No	No	FAIL – risk & margin
200	1700	square	1.09	211.8	325.0	0.43	No	No	FAIL – risk & margin
1250	1700	square	42.46	211.8	250.0	0.43	No	No	FAIL – risk & margin
500	1700	hex	7.85	212.4	390.3	0.42	No	No	FAIL – risk & margin
400	1700	hex	5.02	212.4	382.2	0.42	No	No	FAIL – risk & margin
150	1700	square	0.61	212.4	291.0	0.43	No	No	FAIL – risk & margin
1750	2200	square	49.70	213.2	203.6	0.56	No	No	FAIL – risk & margin
600	1700	hex	11.30	213.2	389.6	0.42	No	No	FAIL – risk & margin
300	1700	hex	2.82	213.3	367.7	0.42	No	No	FAIL – risk & margin
1500	1700	square	61.15	213.4	172.1	0.43	No	No	FAIL – risk & margin
75	2200	square	0.09	213.5	186.2	0.56	No	No	FAIL – risk & margin
2000	2200	hex	74.95	213.6	144.2	0.55	No	No	FAIL – risk & margin
700	1700	hex	15.38	213.8	380.4	0.42	No	No	FAIL – risk & margin

(continued on next page)

(continued from previous page)

N	s	pattern	T(%)	blur(μm)	risk	margin	Safe	Rec	Reason
400	1200	square	8.73	214.3	367.8	0.30	No	No	FAIL – risk & margin
750	1700	hex	17.65	214.4	373.5	0.42	No	No	FAIL – risk & margin
300	1200	square	4.91	214.7	362.2	0.30	No	No	FAIL – risk & margin
100	1700	square	0.27	214.8	230.3	0.43	No	No	FAIL – risk & margin
500	1200	square	13.64	214.9	363.7	0.30	No	No	FAIL – risk & margin
600	1200	square	19.63	215.5	345.5	0.30	No	No	FAIL – risk & margin
700	1200	square	26.73	216.1	317.4	0.30	No	No	FAIL – risk & margin
750	1200	square	30.68	216.2	300.8	0.30	No	No	FAIL – risk & margin
1000	1700	hex	31.38	216.3	320.9	0.42	No	No	FAIL – risk & margin
200	1700	hex	1.26	216.9	325.0	0.42	No	No	FAIL – risk & margin
200	1200	square	2.18	217.0	323.5	0.30	No	No	FAIL – risk & margin
1250	1700	hex	49.03	217.1	250.0	0.42	No	No	FAIL – risk & margin
1000	1200	square	54.54	217.2	206.1	0.30	No	No	FAIL – risk & margin
1000	1200	hex	62.98	217.6	206.1	0.30	No	No	FAIL – risk & margin
2000	2200	square	64.91	217.8	146.4	0.56	No	No	FAIL – risk & margin
150	1700	hex	0.71	218.2	291.0	0.42	No	No	FAIL – risk & margin
1500	1700	hex	70.61	218.3	172.1	0.42	No	No	FAIL – risk & margin
75	1700	square	0.15	218.5	170.1	0.43	No	No	FAIL – risk & margin
600	700	square	57.70	218.8	205.0	0.18	No	No	FAIL – risk & margin
600	700	hex	66.63	219.1	205.0	0.17	No	No	FAIL – risk & margin
150	1200	square	1.23	219.2	276.1	0.30	No	No	FAIL – risk & margin
500	700	square	40.07	219.3	259.9	0.18	No	No	FAIL – risk & margin
500	700	hex	46.27	219.6	259.9	0.17	No	No	FAIL – risk & margin
400	700	square	25.65	219.7	295.3	0.18	No	No	FAIL – risk & margin
750	1200	hex	35.43	219.9	300.8	0.30	No	No	FAIL – risk & margin
300	700	square	14.43	220.1	296.0	0.18	No	No	FAIL – risk & margin
400	700	hex	29.61	220.3	295.3	0.17	No	No	FAIL – risk & margin
700	1200	hex	30.86	220.6	317.4	0.30	No	No	FAIL – risk & margin
300	700	hex	16.66	220.8	296.0	0.17	No	No	FAIL – risk & margin
200	700	square	6.41	221.2	232.2	0.18	No	No	FAIL – risk & margin
200	700	hex	7.40	221.5	232.2	0.17	No	No	FAIL – risk & margin
600	1200	hex	22.67	222.0	345.5	0.30	No	No	FAIL – risk & margin
100	1700	hex	0.31	222.1	230.3	0.42	No	No	FAIL – risk & margin
150	700	hex	4.16	222.9	153.1	0.17	No	No	FAIL – risk & margin
150	700	square	3.61	222.9	153.1	0.18	No	No	FAIL – risk & margin
100	1200	square	0.55	222.9	181.8	0.30	No	No	FAIL – risk & margin
500	1200	hex	15.74	224.0	363.7	0.30	No	No	FAIL – risk & margin
150	1200	hex	1.42	224.6	276.1	0.30	No	No	FAIL – risk & margin
200	1200	hex	2.52	225.0	323.5	0.30	No	No	FAIL – risk & margin

(continued on next page)

(continued from previous page)

N	s	pattern	T(%)	blur(μm)	risk	margin	Safe	Rec	Reason
100	1200	hex	0.63	225.1	181.8	0.30	No	No	FAIL – risk & margin
400	1200	hex	10.08	225.2	367.8	0.30	No	No	FAIL – risk & margin
300	1200	hex	5.67	225.4	362.2	0.30	No	No	FAIL – risk & margin
100	700	hex	1.85	225.8	23.9	0.17	No	No	FAIL – risk & margin
100	700	square	1.60	225.9	23.9	0.18	No	No	FAIL – risk & margin
75	1700	hex	0.18	225.9	170.1	0.42	No	No	FAIL – risk & margin
75	1200	square	0.31	226.9	94.9	0.30	No	No	FAIL – risk & margin
75	1200	hex	0.35	227.3	94.9	0.30	No	No	FAIL – risk & margin
75	700	square	0.90	229.0	0.0	0.18	No	No	FAIL – pitch margin
75	700	hex	1.04	229.1	0.0	0.17	No	No	FAIL – pitch margin